

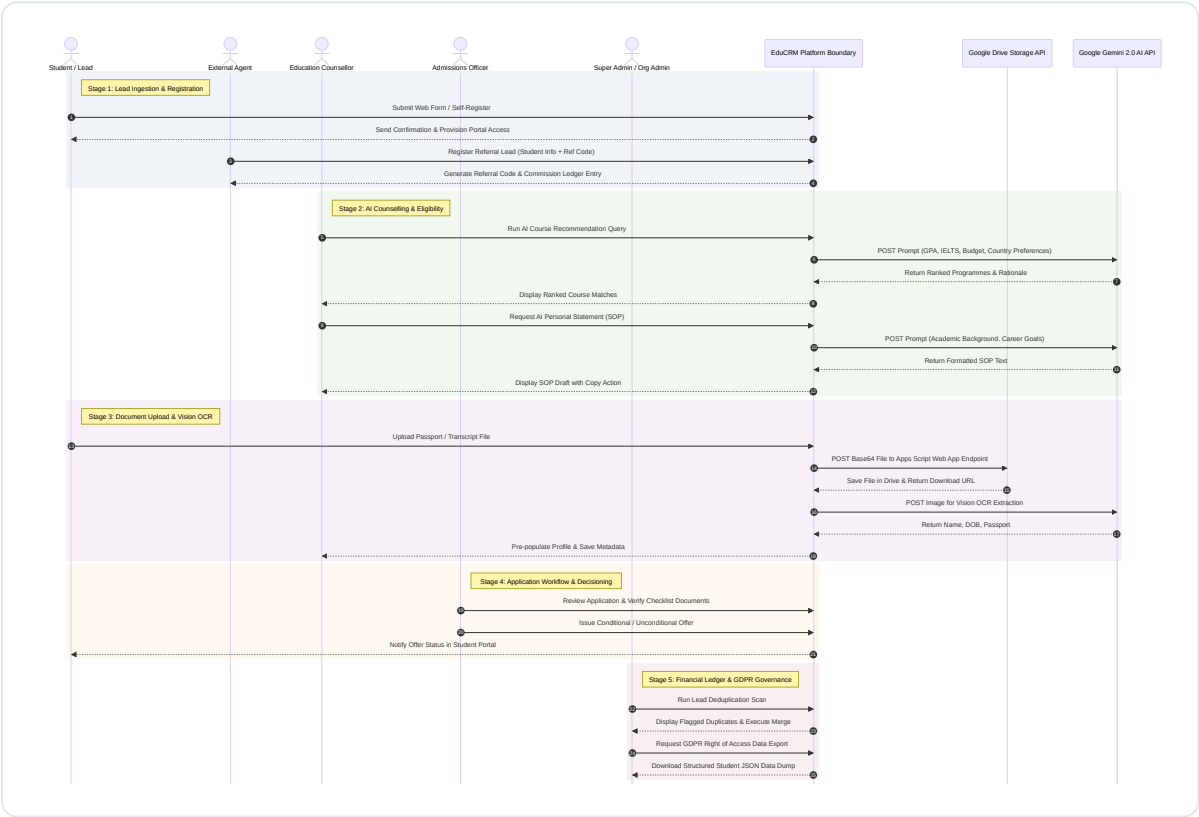
EduCRM — Complete System & Component Sequence Diagrams (UML)

Comprehensive Architectural Specification & Plain-White UML Sequence Diagrams (CRM.pdf)

SYSTEM VERSION	STORAGE ENGINE	AI INTELLIGENCE	SECURITY TIER
v2.4.0 Production	Google Drive API (Apps Script)	Gemini 2.0 Flash REST API	Spark Free Tier (0 Cost)

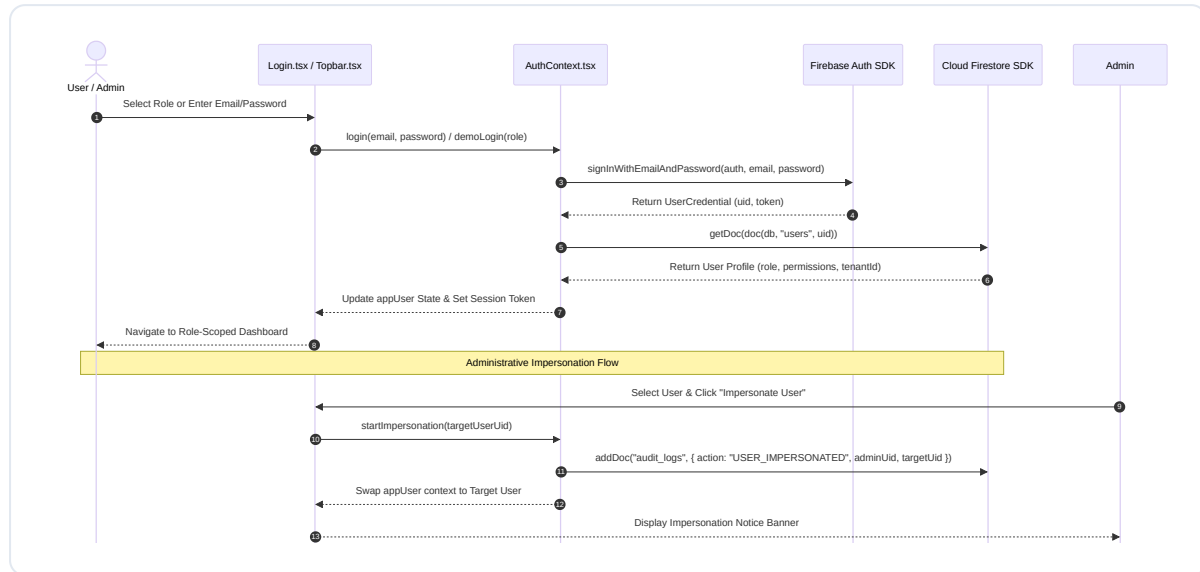
SSD System Sequence Diagram — High-Level Boundary

High-level interactions between external actors (Student, Agent, Counsellor, Admissions, SuperAdmin) and the EduCRM platform boundary.



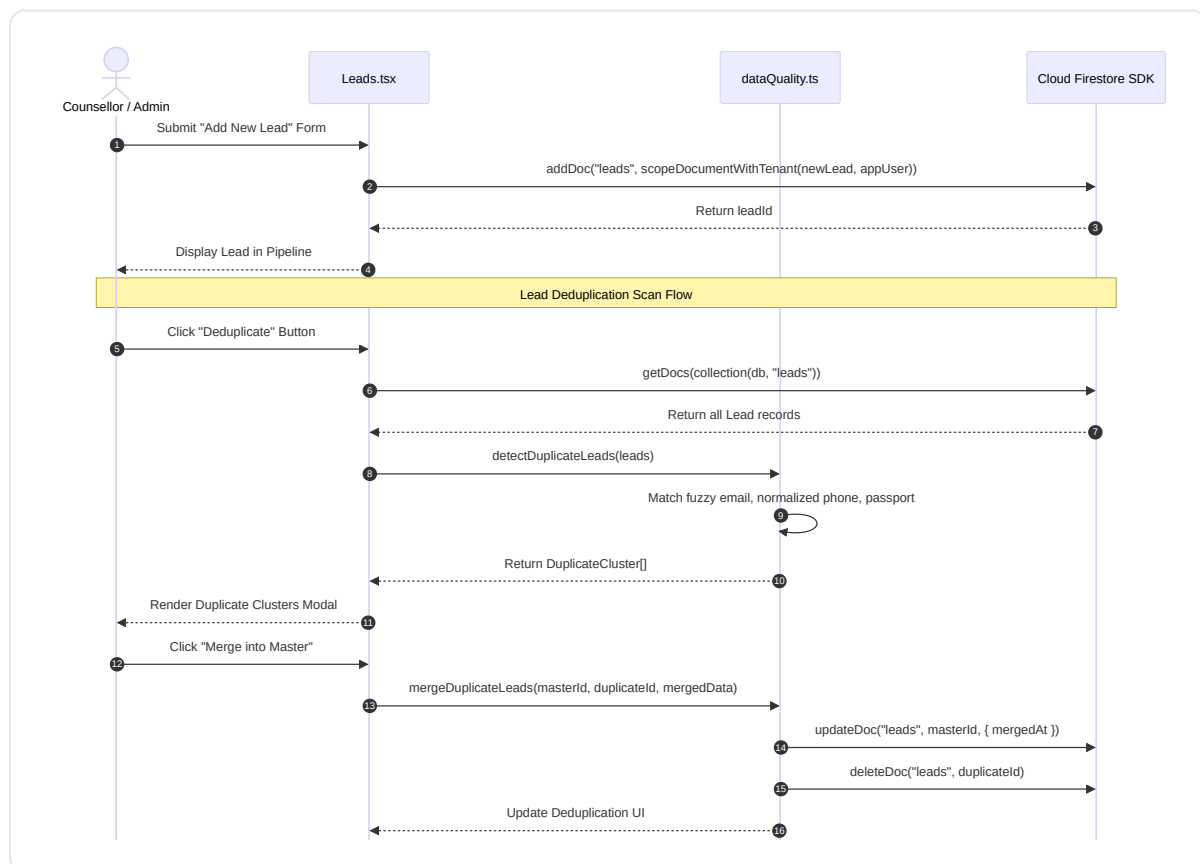
SD-1 Authentication, Session Management & User Impersonation

User registration, email/password & demo role login, session timeout enforcement, and administrative user impersonation with audit logging.



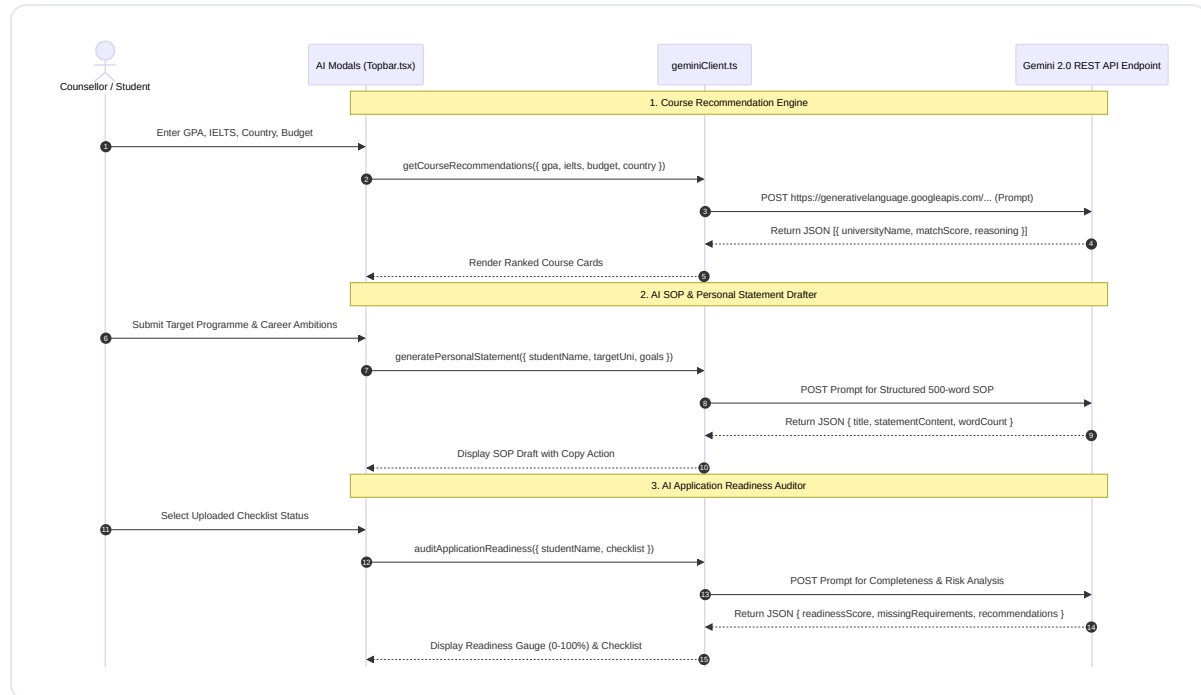
SD-2 Lead Lifecycle Management & Deduplication Engine

Lead creation, stage updates, fuzzy duplicate detection across email, phone, and passport, and master lead merging.



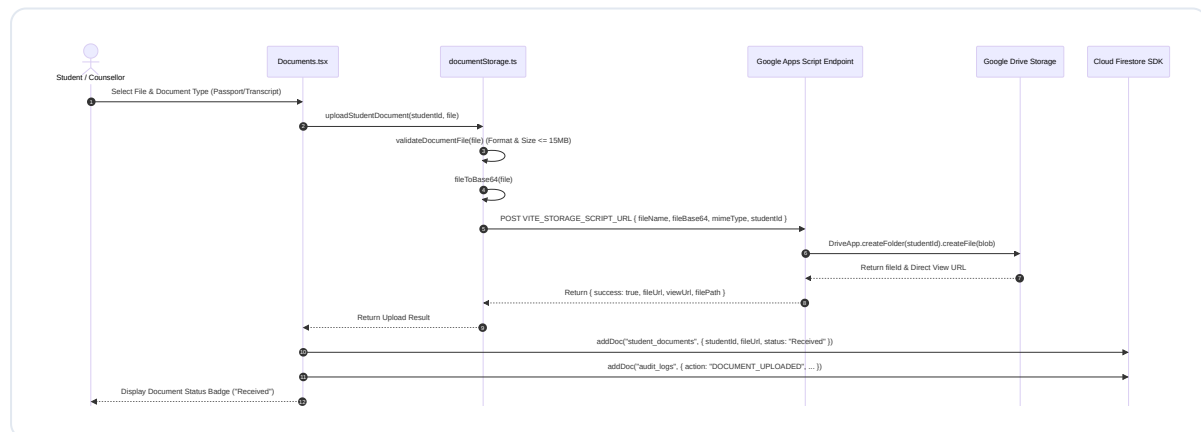
SD-3 Gemini 2.0 AI Counsellor Suite Execution

Client-side integration calling Google Gemini 2.0 Flash for Course Recommendation, Visa Probability Risk, OCR Document Vision, SOP Drafting, and Readiness Audits.



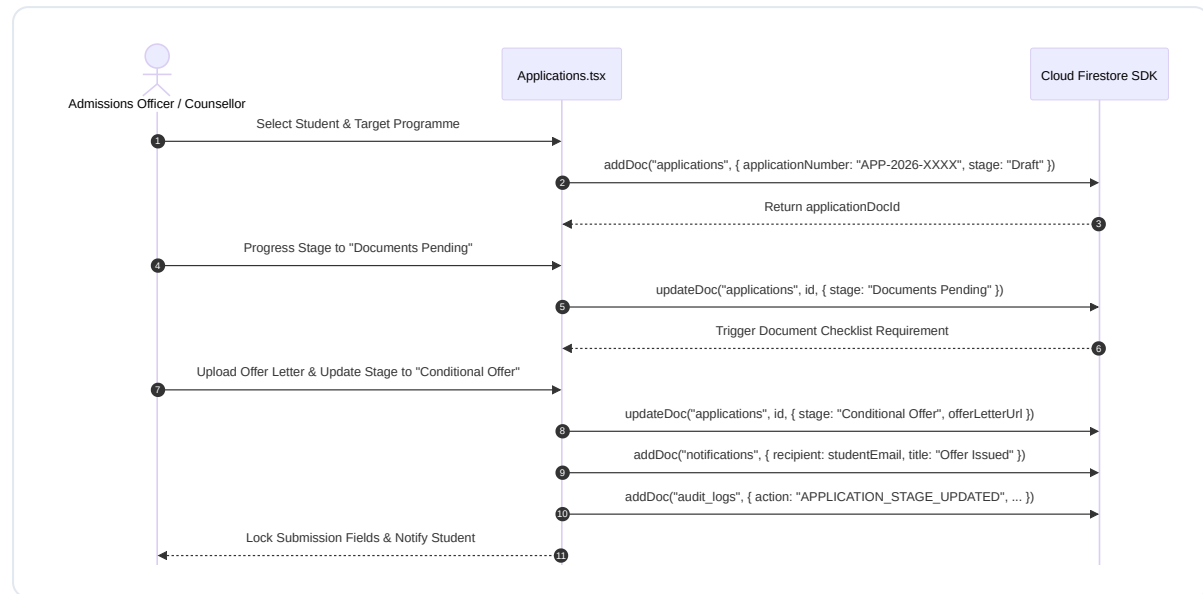
SD-4 Zero-Cost Document Upload & Verification (Google Drive)

Client-side validation, Base64 conversion, upload to Google Apps Script Web App endpoint, Google Drive storage, and metadata persistence in Firestore.



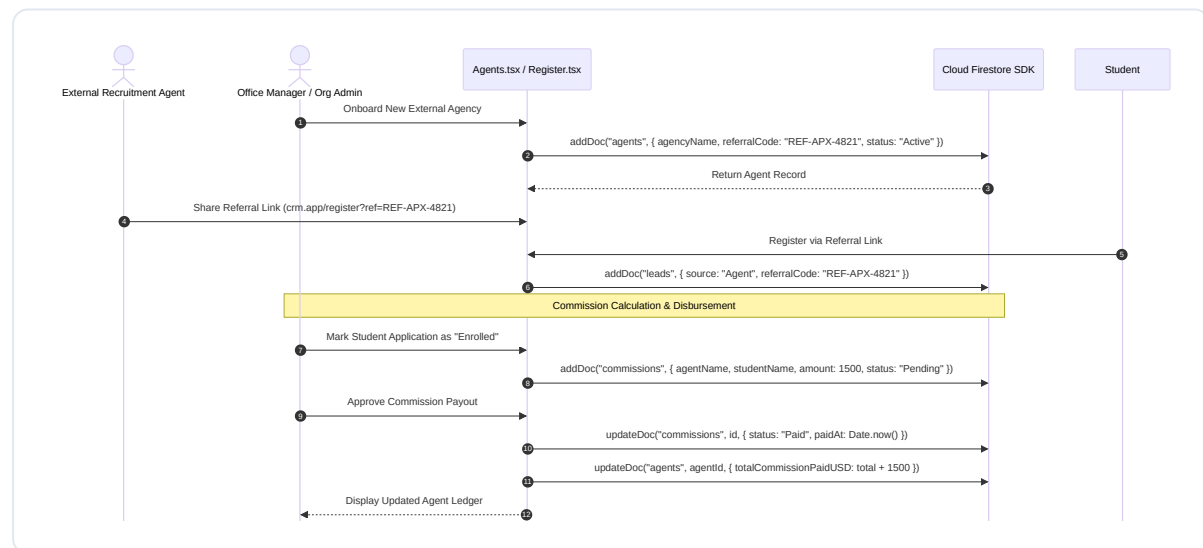
SD-5 20-Stage Application Pipeline & Admissions Decisioning

Application creation, 20-stage status progression, required document checklists, conditional/unconditional offer generation, and submission field locking.



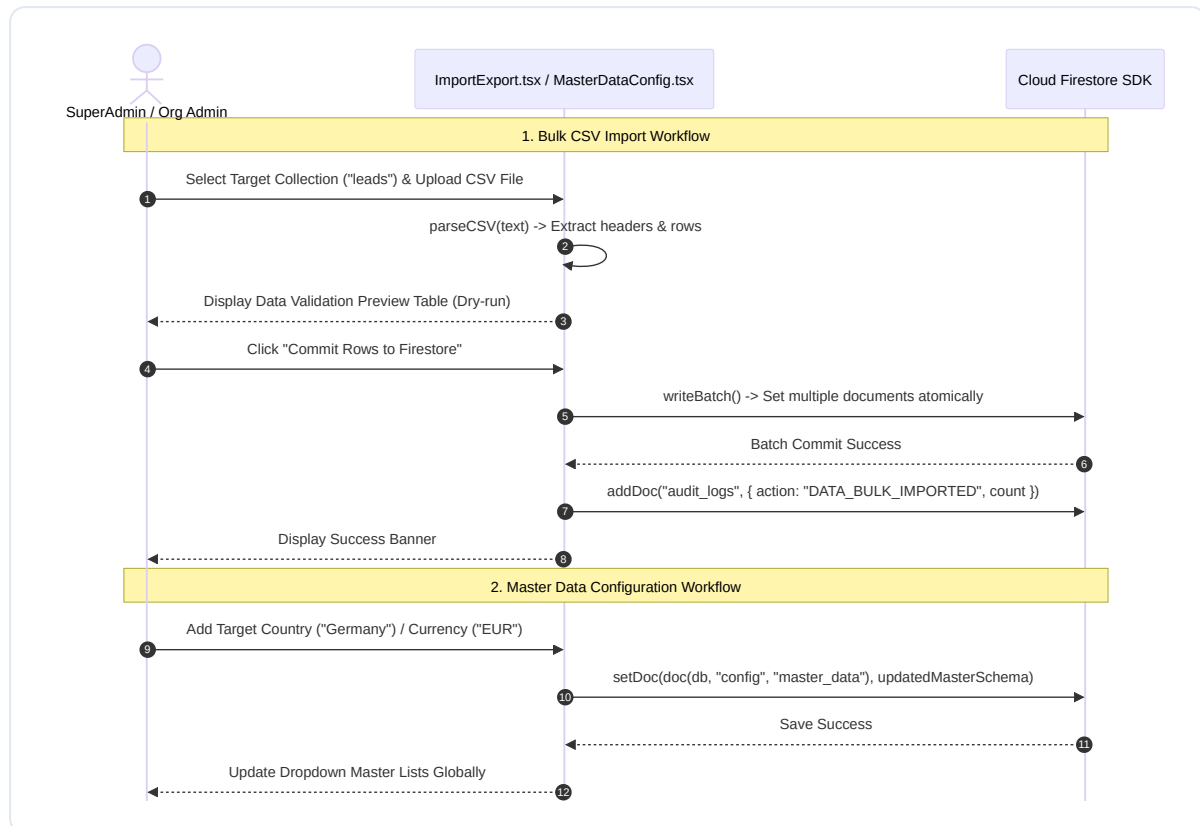
SD-6 External Agent Network, Referral Tracking & Commission Ledger

Agent onboarding, referral code generation, referral link tracking, enrolment confirmation, commission calculation, and disbursement logging.



SD-7 CSV Data Migration & Master Data Configuration Center

Bulk CSV parsing, column mapping preview, dry-run schema validation, Firestore batch writing, and master data dropdown administration.



SD-8 **GDPR Privacy Compliance & Data Quality Audit Engine**

Right of Access structured JSON data exports, Right to be Forgotten PII anonymization, automated data quality rule checking, and natural-language AI reporting.

